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MILITARY SPECIFICATION

CUTOFF MACHINES, ABRASIVE DISK, METAL CUTTING, CHOP-STROKE, WET CUT, FLOOR MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers floor mounted, chop-stroke, wet cut, abrasive disk cutoff machines with equipment and accessories as described herein.

1.2 Classification. This specification covers machines of the following types, classes, and sizes. The type, class, and size machine required for specific procurements shall be as specified (see 6.2.1).

Type I - Chop-stroke, Non-oscillating Head
- Class 1 - Manual wheelarm feed
  - Size 16 - 16-inch disk diameter
  - Size 18 - 18-inch disk diameter
  - Size 20 - 20-inch disk diameter
- Class 2 - Power wheelarm feed
  - Size 16 - 16-inch disk diameter
  - Size 18 - 18-inch disk diameter
  - Size 20 - 20-inch disk diameter
- Class 3 - Automatic machines
  - Size 18 - 18-inch disk diameter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3405

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Type II - Chop-stroke, Oscillating Head
Class 1 - Manual wheelarm feed
Size 18 - 18-inch disk
Size 20 - 20-inch disk
Size 26 - 26-inch disk

Class 2 - Power wheelarm feed
Size 18 - 18-inch disk
Size 20 - 20-inch disk
Size 26 - 26-inch disk
Size 34 - 34-inch disk

Class 3 - Automatic machines
Size 20 - 20-inch disk

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standard for.

ANSI/B7.1 - American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY (10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

ASTM-E-380 - Metric Practice Guide.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.2 Design. The machine shall be one of the manufacturer's current models, new and unused. The machine shall be capable of cutting hardened or annealed ferrous metals, nonferrous metals, plastics, ceramics and refractory materials in the form of solid bars, tubing, structural sections, and extrusions. The design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM-E-380.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

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3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 Components.

3.4.1 Base. The base shall be either a casting or a steel weldment designed for floor mounting. It shall support the workholder, wheelarm, and controls at a convenient operating height. The surface of the base adjacent to the abrasive disk shall have a coolant collection trough or other means for collecting spent coolant and returning it to the coolant reservoir. The base shall have means for leveling and bolting the machine to the floor or foundation.

3.4.2 Wheelarm. The wheelarm shall be either a casting or steel weldment supporting the spindle, spindle motor, spindle drive mechanism, and counterbalance. The wheelarm shall be of the chop-stroke type that pivots about a pivot pin mounted on the machine base. The pivot bearings shall be of the ball or roller type sealed to prevent loss of bearing lubricant and entrance of coolant, metal cuttings, abrasive dust, or other extraneous materials. When specified (see 6.2.1), the wheelarm shall be mounted on a swivel type wheelhead that will enable the machine to cut any angle from 45 degrees left, to 45 degrees right of the workpiece axis.

3.4.2.1 Wheelarm feed system. Class 1 machines shall have a manually operated feed lever attached to the wheelarm at a location convenient to the operator. The wheelarm shall be counterbalanced so that it returns to the raised position when the feed lever is released. Unless otherwise specified (see 6.2.1), class 2 and class 3 machines shall have a hydraulic or pneumatic cylinder that feeds the arm downward during the cut and returns the arm to the raised position upon completion of the cut. The feed force shall be controllable by regulation of air or hydraulic pressure to the cylinder. The power feed system of class 3 machines shall be automatic and shall operate in accordance with the sequence of operations specified in 3.4.11.1.

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3.4.2.2 Wheelarm interlock. All class 2 and class 3 machines shall have mechanical or electrical interlocks that prevent the wheelarm from feeding downward if the workpiece is not securely clamped in the workholder.

3.4.2.3 Wheelarm stop. If the machine design is such that over travel of the wheelarm would allow the abrasive disk to cut into the machine structure, the machine shall have an adjustable stop that limits wheelarm travel and prevents such damage.

3.4.3 Spindle. The spindle shall be mounted on the wheelarm and shall rotate on antifriction bearings of a size and type suitable for the application. The bearings, or bearing housings, shall be sealed to prevent lubricant leakage or entrance of coolant, metal cuttings, abrasive dust, or other extraneous materials. The spindle shall have a pair of disk flanges of a size conforming to the requirements of ANSI B7.1. The means for attaching the abrasive disk to the spindle shall be such that full drive motor torque can be transmitted to the disk without causing disk slippage on the spindle.

3.4.4 Spindle oscillation mechanism (type II machines only). All type II machines shall have a mechanism that oscillates (reciprocates) the abrasive disk in a direction parallel to the kerf in the workpiece. Unless otherwise specified (see 6.2.1), the mechanism shall be powered either by an electric motor having a horsepower rating conforming to the requirements of table I, or by a hydraulic or pneumatic cylinder. Unless otherwise specified (see 6.2.1), the length of the oscillation stroke shall be infinitely variable over a range of 0 to 1, or more, inches on sizes 18 and 20 machines, and over a range of 0 to 4, or more, inches on sizes 26 and 34 machines.

3.4.5 Spindle drive system. The spindle shall be directly driven by the drive motor or through a timing belt, multiple belt or polybelt drive system. If a belt drive system is employed, the system shall have means for adjusting and maintaining proper tension of the drive belts. Belt drive systems shall be capable of transmitting full drive motor torque without belt slippage. The drive system shall drive the abrasive disk at a speed in the range of 6,500 to 11,000 surface feet per minute, measured at the periphery of a disk having the diameter specified in table I.

3.4.6 Abrasive disk. Abrasive disks shall be available from the machine manufacturer for use on the machine for cutting hardened or annealed ferrous metals, nonferrous metals, plastics, ceramics and refractory materials. When required (see 6.2.1), the specified quantity and type of abrasive disks shall be furnished with the machine.

3.4.7 Workholder. All machines shall have a workholder that will hold the workpiece in proper position for a cut 90 degrees to the workpiece axis. When specified (see 6.2.1), class 1 and class 2 machines shall have a swivel type workholder that will enable the machine to cut any angle from 45 degrees left, to 45 degrees right, of the workpiece axis. The workholder

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may be of the vise, fence, V-block, or L-block type. The clamping device for class 1 and class 2 machines shall be either manually or power operated, either by hand or foot. When specified (see 6.2.1), the clamping device for class 1 and class 2 machines shall be either manually or power operated in accordance with the sequence of operation requirements in 3.4.11.1.

3.4.8 Wheel guards. The machine shall have wheel guards that protect the operator from sparks, abraded materials, and coolant splash. The guards shall have sufficient strength to prevent operator injury in the event of disk breakage. The guards shall not interfere with operation of the machine, and shall be removable to facilitate cleaning and servicing of the machine.

3.4.9 Coolant system. All machines shall have a recirculating type coolant system consisting of a sump or reservoir, a power driven pump, and all necessary piping. The reservoir capacity and pump flow rate shall be adequate to dissipate heat generated by the cutting process while cutting at the maximum cutting load. The coolant system shall have means for draining and cleaning the system and shall include strainers or baffles that will separate solid materials from the coolant. The coolant shall be discharged uniformly over both sides of the abrasive disk and the system shall have means for controlling the coolant flow rate. Means shall be provided for collecting spent coolant and returning it to the reservoir for reuse.

3.4.10 Length gauge. Class 1 and class 2 machines shall have an adjustable length gauge to facilitate making repeated cuts to the same part length. Unless otherwise specified (see 6.2.1), the gauge shall be adjustable to cut lengths from 1/2 inch, or less, to 15 inches, or more. The length gauge shall not interfere with operation of the machine, shall present no safety hazard, and shall be readily removed from the machine when not needed.

3.4.11 Automatic machines (class 3 machines only). Class 3 machines shall have an automatic bar feeder, an automatically operated powered workholder, an automatically operated wheelarm feed system and all necessary controls and mechanisms required to effect the automatic sequence of operations specified in 3.4.11.1. The bar feeder shall be capable of accommodating bar stock of a size up to the capacity shown in table I, and having a length of 10, or more, feet. The machine shall be capable of automatically cutting bars into sections having lengths of not less than 0 to 12 inches for type I, class 3 machines, and not less than 0 to 18 inches for type II, class 3 machines.

3.4.11.1 Automatic sequence of operations (class 3 machines only). Class 3 machines shall be automatic to the extent that after the bar stock is loaded onto the bar feeder and the mechanism is adjusted to cut the stock to the desired section length, activation of the "cycle start" button shall cause the bar feeder to automatically feed the stock to the preselected length, the workholder shall automatically grip the stock, and the wheelarm

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shall automatically feed downward to cut the workpiece. After completion of the cut, the wheelarm shall automatically return to the raised position, the workholder clamp shall release and the above described automatic cycle shall repeat until the bar feeder has depleted its stock. After stock depletion, the bar feeder, wheelarm, and workholder shall return to positions suitable for reloading more bar stock into the feeder and the machine shall automatically stop. Means for stopping the machine at any time during the cycle shall be provided.

3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.12.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.12.2 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar aging devices is permitted only for voltage surge protection to other solid state components.

3.4.13 Hydraulic system. The hydraulic system shall be complete and shall include all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump, and on the low side of each pressure reducing valve to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.14 Pneumatic system. The pneumatic system, when utilized, shall be capable of operating on a 90 to 110 pounds per square inch shop air supply system. The system shall consist of an air filter, pressure regulator and necessary pressure gauges, operating valves, relief valves, piping, hoses, and fittings.

3.5 Sizes and capacities. The sizes and capacities of the machine and components shall meet, or exceed, the requirements of table I.

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TABLE I. Sizes and capacities.

MACHINE CHARACTERISTIC	TYPE I, CHOP-STROKE NON-OSCILLATING HEAD						
	Class 1 Manual Wheelarm Feed			Class 2 Power Wheelarm Feed			Class 3 Automatic Machines
	Size 16	Size 18	Size 20	Size 16	Size 18	Size 20	Size 18
Capacity (Ferrous, round), inches							
Solid	2	2-1/2	3	2	2-1/2	3	2
Tubing	4	4-1/2	5	4	4-1/2	5	4
Motor horsepower:							
Disk Drive	10	10	20	10	10	20	10
Coolant pump	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Wheel size, inches:							
Diameter	16	18	20	16	18	20	18
Thickness	1/16 to 1/8	1/16 to 1/8	1/8 to 3/16	1/16 to 1/8	1/16 to 1/8	1/8 to 3/16	1/16 to 1/8
Arbor hole diameter	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2
Flange diameter	←			See 3.4.3			→

MACHINE CHARACTERISTIC	TYPE II, CHOP-STROKE OSCILLATING HEAD							
	Class 1 Manual Wheelarm Feed			Class 2 Power Wheelarm Feed				Class 3 Automatic Machines
	Size 18	Size 20	Size 26	Size 18	Size 20	Size 26	Size 34	Size 20
Capacity (Ferrous, round), inches								
Solid	3	4	6	3	4	6	10	6
Tubing	5-1/2	6	8	5-1/2	6	8	10	6
Motor horsepower:								
Disk Drive	10	15	25	10	15	25	40	15
Coolant pump	1/4	1/4	1/4	1/4	1/4	1/4	3/4	1/4
Oscillator	1/3	1/3	1/3	1/3	1	1	1-1/2	1
Wheel size, inches:								
Diameter	18	20	26	18	20	26	34	20
Thickness	1/16 to 1/8	1/8 to 3/16	1/8 to 3/16	1/8	1/8	5/32	7/32	1/8
Arbor hole diameter	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2	1 or 1-1/2
Flange diameter	←			See 3.4.3				→

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.6 Performance. When tested in accordance with 4.6.2, the machine shall be capable of cutting SAE 1045 steel at a rate of 15 square inches per minute, or more. The cut surfaces shall be not rougher than 40 microinches and shall be square with the axis of the bar within ±0.010 inch per inch of diameter. The equal length pieces shall vary in length not more than ±0.015 inch.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished with the machine by the manufacturer.

b. One abrasive disk (the disk used in performance testing of the machine).

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.10 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.11 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.12 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captains listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
Abrasive disk diameter
Work piece capacity: ____ inches, solid, ferrous, round
____ inches, tubing, ferrous, round

3.13 Technical data. When technical data is required it should be furnished in accordance with the requirements of the contract.

3.14 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and all the tests in 4.6. Failure of the item to pass the examination or tests shall be cause for rejection.

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4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements specified herein.

4.6 Tests.

4.6.1 Operational test. Class 1 and class 2 machines shall be operated for not less than 15 minutes; class 3 machines shall be operated for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period. The automatic operating sequence of class 3 machines shall be tested for conformance with the sequence of operation requirements of 3.4.11.1.

4.6.2 Performance and accuracy test.

4.6.2.1 Preparation for test. All tests shall be performed using an SAE 1020 solid steel bar, in the "as-drawn" or "as rolled" condition, having a nominal diameter equal to the maximum rated capacity of the machine and using a disk conforming to the requirements of table I.

4.6.2.2 Cutting rate tests. Class 1 and class 2 machines shall cut three disks, not less than one-inch thick, from the workpiece. Class 3 machines shall cut five disks, not less than one-inch thick, from the workpiece using the automatic mechanism. All cuts shall be at a 90 degree angle to the workpiece axis. The actual cutting time shall be observed and the cutting rate shall be calculated. The cutting rate for all machines shall be not less than 15 square inches per minute. There shall be no evidence of damage (wear excepted) to the abrasive disk.

4.6.2.3 Accuracy test. The last cut in the test specified in 4.6.2.2 shall produce a piece not less than 4 inches long. The cuts on the 4-inch piece shall be square with the axis of the workpiece within $\pm$0.010 inch per inch of diameter. All cuts shall have a surface finish not rougher than 40 microinches. The equal length pieces shall vary in length not more than $\pm$0.015 inch.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. Machines covered by this specification are intended for cutting off lengths of hardened or annealed ferrous metals, nonferrous metals, plastics, ceramics, and refractory materials in the form of solid bars, tubing, structural sections, and extrusions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Swivel type wheelarm, if required (see 3.4.2).

j. Wheelarm feed system for class 2 and class 3 machines, if different (see 3.4.2.1).

k. Spindle oscillation motor horsepower, if different (see 3.4.4 and table I).

l. Spindle oscillation mechanism stroke, speed selections and length, if different (see 3.4.4).

m. Quantity and type of abrasive disks, if required (see 3.4.6).

n. Swivel type workholder for class 1 and class 2 machines, if required (see 3.4.7).

o. Power operated workpiece clamping device for class 1 and class 2 machines, if required (see 3.4.7).

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p. Length gauge, if different (see 3.4.10).

q. Electrical system and input power supply, if different (see 3.4.12).

r. Motor enclosure, if different (see 3.4.12.1).

s. Standar' equipment, if different (see 3.7).

t. Specify and fully describe optional equipment required, if different (see 3.8).

u. Specify applicable repair parts, if required (see 3.9).

v. Lubrication chart or plate, if different (see 3.11).

w. Nameplate, if different (see 3.12).

x. First article inspection, if different (see 4.3).

y. Quality conformace inspection, if different (see 4.4).

z. Level of packaging, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements, such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard

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cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - SM
Navy - YD, MC

Preparing Activity:
DLA-IP

Project Number:
3405-0075

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-80071B	Cutoff Machines, Abrasive Disk, Metal Cutting, Chop-Stroke. Wet Cut, Floor Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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